Rating Players from Small Leagues: A Walk-Forward Test on the Belize Domestic League, 2024–26

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Independent analysis · designed and directed, built with AI coding tools · August 2026. Revised October 22, 2026, after an independent data validation. Player data anonymized; no individual ratings published. Every number below is produced by the project's code and listed in its `results/key_numbers.csv`.

Abstract

Players from small leagues are hard to judge: their seasons are short, their box scores are partial, and no official exchange rate links their competition to NCAA Division I. This project places box-score lines from the Belize domestic league, three FIBA youth and regional tournaments and Spain's third division (Segunda FEB) on an estimated NCAA Division I-equivalent box-score (BPM) scale, through translation offsets estimated from players seen in two competitions, and shrinks each player's short seasons toward a prior. Only the domestic prediction was tested. In a walk-forward test on the Belize domestic league, 102 player-seasons from 55 players, 2024–26, the shrunk rating predicted next season's composite with RMSE 2.24 vs 3.35 for raw prior-season averages (lower is better). Against the average of covered players (2.60) the gain was 0.36 BPM (95% CI about −0.1 to +0.9), not distinguishable from zero, and a plain 50/50 blend scored 2.28. Most of the gain over raw averages is regression to the mean: a short season should not be taken at face value. The model's predictions also run low by 1.29 BPM on average, and the translation offsets rest on 4 to 19 players each and were not tested. The test is exploratory, not pre-registered.

1. Question

How much should a team trust a box-score line from a small league? In the Belize domestic league the median player-season runs 176 to 203 minutes, depending on the season; the published box scores cover 58–93% of each team's minutes; and no official translation to NCAA Division I exists. The project asks two questions. Can lines from different competitions be put on one estimated scale? And does shrinking a player's short seasons toward a prior predict his next season better than his raw numbers or simple averages? Only the second was tested out of sample, and only for domestic players predicting their next domestic season. The work is box-score analysis only: it rates no team, picks no squad and publishes no player's rating.

2. Data and design

Input	Source	Size
Belize domestic league (BEBL) box scores, 2023–2026	basketball.latinbasket.com (third-party aggregator)	303 lines, 297 player-seasons; 30 team-seasons
FIBA U17 Centrobasket 2023; FIBA U18 AmeriCup 2024	fiba.basketball	94 and 96 player lines
2026 CAC Games, men's basketball	the Games' official results site	94 player lines
Spain's third division (Segunda FEB)	FEB live stats	recorded lines of 7 bridge players
NCAA D1 player panel	Torvik (barttorvik.com), frozen copy	90,739 player-seasons; 40,822 fit the transfer

Names were replaced by random keys in a private registry before any analysis, and tables are joined on keys, never on names. The repository, its figures and this paper contain only aggregates over cells of five or more players; some of the tournament players are minors. In 20 of the 30 domestic team-seasons the box scores record under 180 minutes a game, so part of every team is missing from the tables.

Unit	One Belize domestic player-season
Observations	102 test player-seasons from 55 players (30/37/35 in 2024, 2025 and 2026), each with 150+ minutes in the target season and an earlier covered season
Outcome	Next season's box-score composite on the estimated D1-equivalent BPM scale; scored by RMSE, lower is better
Baselines	Average of covered players; raw prior-season average; last prior season; 50/50 blend
Evidence type	Exploratory walk-forward, not pre-registered; two-sided decision rule at $\alpha = 0.05$, set after the validation rerun

3. Methods

3.1 From a box line to the scale

Each box line is converted to an estimated BPM (box plus/minus: points per 100 possessions above the D1 average, as Torvik defines it) by a linear transfer fit on D1 player-seasons. The domestic tables have no turnovers or offensive rebounds, so domestic lines use a "split" lane: points, rebounds, assists, steals and blocks per 40 minutes, three shooting percentages and minutes per game. Its cross-validated R^2 is 0.494, against 0.669 for the "full" lane used for complete tournament box scores; folds grouped by player give the same values. Shooting percentages are shrunk toward their pool by attempts (a beta-binomial fit), so a 3-for-4 week does not count as 75% shooting. The result for one player-season is its composite.

3.2 The rating

A player's earlier seasons are combined with weights that grow with minutes, then shrunk toward one D1 prior: low-major wing guards playing 15–22 minutes a game (mean −2.90 BPM, SD 2.91). The D1 reliability weight of a season of m minutes is $m/(m + k)$ with $k = 153$ minutes, so a 153-minute season gets half weight and no season gets full weight. The chance SD of one season's composite is 4.6 BPM at 100 minutes and 2.7 at 300. In the tested model the weight also counts the transfer's own error, so one season of 153 minutes gets 36%.

3.3 The walk-forward test

Each target season (2024, 2025, 2026) is predicted from earlier seasons only, and every constant, the shooting shrinkage included, is refit on those earlier seasons; the code asserts that no input is dated after the target. The model is compared with four baselines:

- **Average of covered players:** the minutes-weighted mean composite of every player-season the domestic box scores cover before the target season. The tables cover only 58–93% of team minutes, so it averages the players the tables include.
- **Raw prior-season average:** the minutes-weighted mean of the player's own earlier composites, unshrunk.
- **Last prior season:** his most recent earlier composite.
- **50/50 blend:** half the raw prior-season average plus half the average of covered players.

Scores are RMSE in BPM (the typical size of a miss; lower is better), mean and median absolute error, and Spearman rank correlation. A gain is a baseline's RMSE minus the model's, so a positive gain means the model is better.

3.4 Uncertainty and the decision rule

Intervals come from a paired bootstrap that resamples whole players (2,000 draws), because the 102 rows come from 55 players and one player's seasons are not independent. Across ten bootstrap seeds the interval ends move by up to 0.12 BPM, so the text quotes them to one decimal. Each comparison carries a two-sided bootstrap p-value from the same draws, judged at $\alpha = 0.05$. That rule was written after the validation rerun, so the test is exploratory, not pre-registered. A table of 21 sensitivity specifications was declared in the configuration file before it ran.

4. Results

4.1 Next-season prediction

Predictor	RMSE	95% CI	Median abs. error	MAE	Spearman
Model (shrunk rating)	2.24	1.9 to 2.6	1.54	1.81	0.675
Average of covered players	2.60	2.2 to 3.0	1.72	2.09	n/a
Raw prior-season average	3.35	2.4 to 4.4	1.66	2.32	0.645
Last prior season	3.39	2.6 to 4.3	1.82	2.43	0.623
50/50 blend	2.28	1.9 to 2.6	1.41	1.73	0.634

Table 1. 102 test player-seasons from 55 players, target seasons 2024–26; errors in BPM, lower is better. Intervals resample whole players; their ends are good to about one decimal. The average of covered players gives everyone in a season the same prediction, so it has no meaningful rank correlation.

Model compared with	Gain (BPM)	95% CI	Bootstrap p	Decision at α = 0.05
Raw prior-season average	+1.11	+0.1 to +2.2	0.01	H0 rejected
Last prior season	+1.15	+0.3 to +2.1	<0.01	H0 rejected
Average of covered players	+0.36	−0.1 to +0.9	0.13	H0 not rejected
50/50 blend	+0.04	−0.4 to +0.5	0.88	H0 not rejected

Table 2. Gain = baseline RMSE minus model RMSE; positive means the model is better. H0: the two RMSEs are equal (two-sided).

Against raw prior-season averages the claim is supported: the model's typical miss is about a third smaller, 2.24 against 3.35 BPM, a gain of +1.11 (95% CI about +0.1 to +2.2). Against the average of covered players the result is inconclusive: the gain is +0.36 BPM (95% CI about −0.1 to +0.9), so the data neither show a gain nor rule out one up to about +0.9 BPM, and this is not evidence of no gain. Near the D1 median, 0.36 BPM is worth about 3 percentile points, while the model's RMSE is about 20. The 50/50 blend nearly matches the model (RMSE 2.28) and has the smaller median absolute error (1.41 against 1.54). Rank correlations do not separate the model from raw averages: the difference is +0.031 (95% CI −0.03 to +0.10). Most of the gain over raw averages comes from not taking short seasons at face value, which the blend also does (Figure 1).

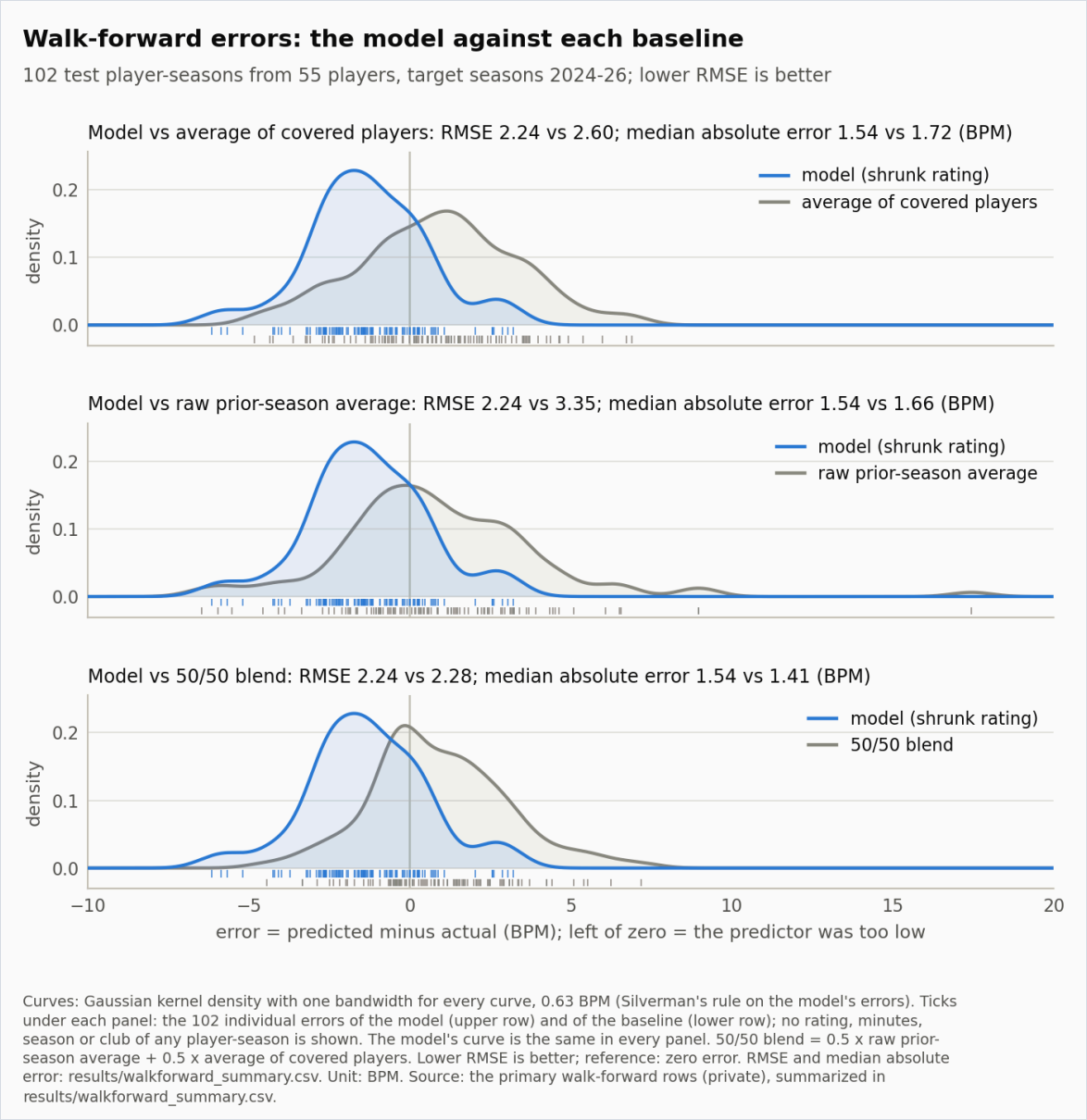


Figure 1. Walk-forward errors (predicted minus actual) of the model against each baseline: kernel densities with one shared bandwidth, and the 102 individual errors as ticks under each panel. Left of zero, the prediction was too low.

4.2 How stable the gain is

Twenty-one specifications, each changing one choice of the primary row and recomputed inside the folds, were declared before they ran. Against raw prior-season averages the model has the lower RMSE in 20 of 21, and the interval excludes zero in 15 of 21. The exception is a 400-minute floor: on the 25 test player-seasons with 400+ minutes, the gain over the raw prior-season average is −0.53 (95% CI about −1.0 to −0.1), so for players with plenty of minutes, shrinking toward the prior costs more than it saves.

Against the average of covered players the gain is positive in 18 of 21, but its interval excludes zero in only 3 of 21. It falls as the minutes floor rises, from +0.58 at 100 minutes to −0.16 at 250 and −0.36

at 400 (Figure 2). It is +0.00 without the 2024 season, and −0.06 when the transfer's full residual SD of 2.88 BPM is counted as noise. Leaving out any one player keeps it between +0.25 and +0.48. A state-space version of the rating, which lets ability drift between seasons, scores about the same as the tested estimator; it is a sensitivity row, not the tested model.

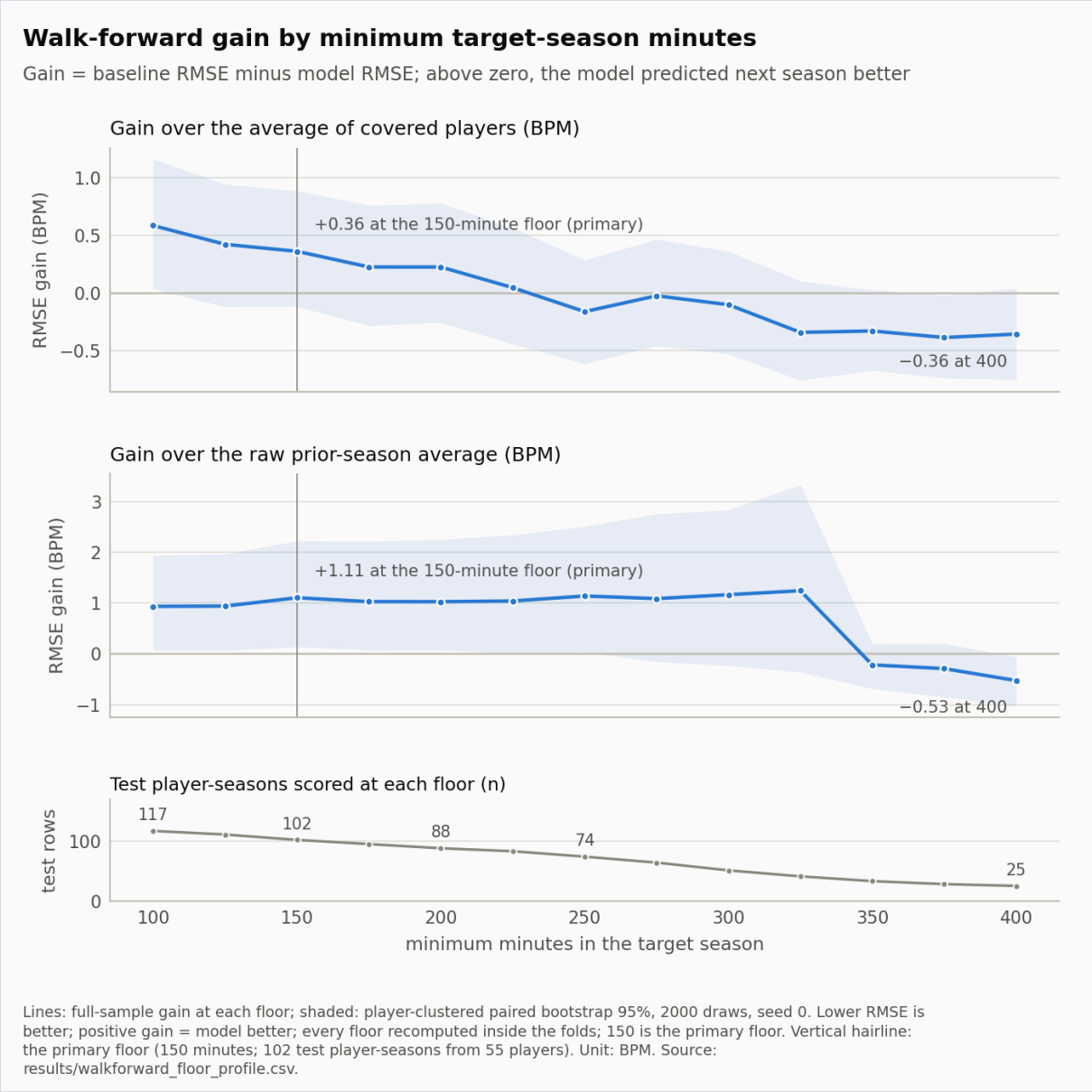


Figure 2. Gain by minimum target-season minutes, each floor recomputed inside the folds (player-clustered paired bootstrap 95% bands). The vertical line marks the primary 150-minute floor; the bottom panel shows how many test player-seasons each floor keeps.

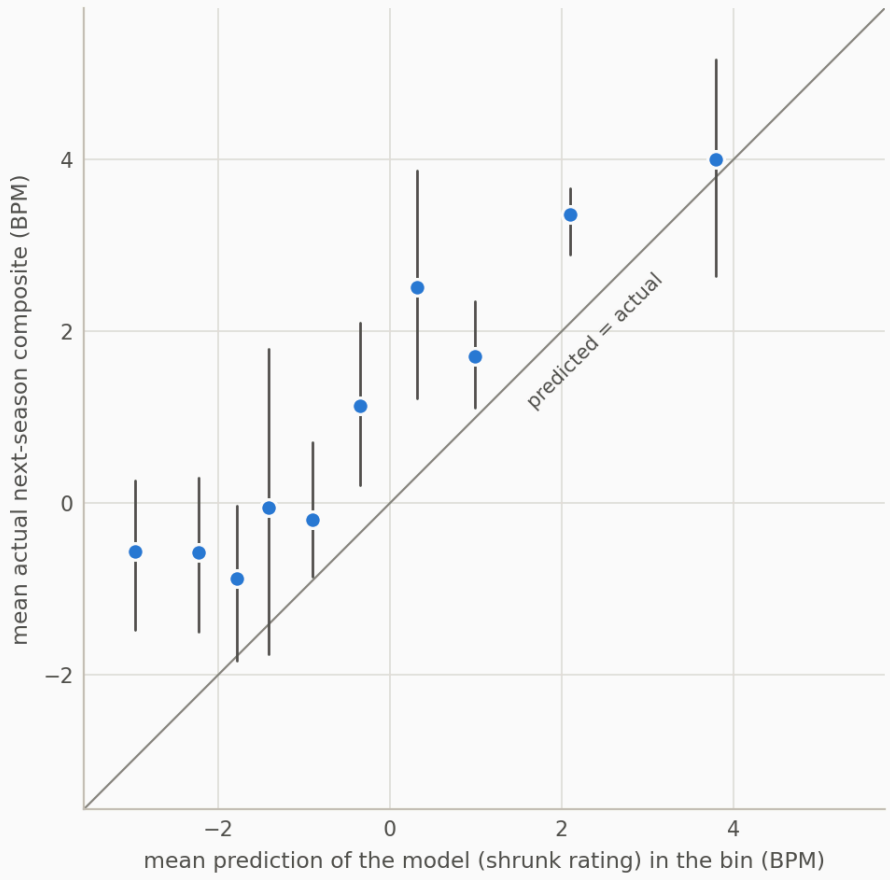
4.3 The model runs low

The model's predictions are too low by 1.29 BPM on average (95% CI about +0.9 to +1.6), and 78 of 102 misses point the same way. The bias grows with target-season minutes: +0.24 at 150–249 minutes, +1.72 at 250–399 and +1.62 at 400 or more. The cause is centering. Every domestic player is shrunk toward one D1 group whose mean is −2.90, while the outcome stays on the domestic

composite, whose mean is +1.06, and a typical prediction gives the player's own seasons only about half the weight (mean 0.53). A prior centered on all of D1 would score RMSE 1.98, but it was found after looking at the test seasons, so it is reported as post hoc, not adopted, and can only be tested on the 2027 season (Figure 3).

Binned calibration: actual vs predicted

10 bins of the model's predictions, test rows 2024-26; above the diagonal = under-prediction



Dots: bin means (10-11 test player-seasons and 6+ players per bin; 102 rows from 55 players in all). Whiskers: 95% interval of the bin's mean actual (player-clustered bootstrap, 2000 draws). Reference: the diagonal, predicted = actual. Overall mean residual (actual minus model): +1.29 BPM (results/residuals.csv); the model's RMSE, where lower is better, is in results/walkforward_summary.csv. Unit: BPM. Source: results/wf_binned_calibration.csv.

Figure 3. Actual against predicted next-season composite in ten bins of the model's predictions (10–11 test player-seasons and 6+ players per bin), with player-clustered bootstrap 95% intervals of each bin's mean. Points above the diagonal are under-predictions.

4.4 How sure for one player

Ninety-five percent of the model's misses fall within ±4.2 BPM (signed range −2.7 to +5.4, shifted by the bias above). The model's own next-season band is too wide: it covers 90%, 96% and 99% of outcomes at nominal 68%, 80% and 95%. The domestic composite is moderately stable: its season-

to-season correlation is 0.62 over 70 pairs of seasons with 150+ minutes each, close to the 0.65 that D1 reliability implies. Every interval reflects sampling noise only, not who is in the sample.

4.5 Translation offsets (estimated, not tested)

A translation offset is the average change in rating for players seen in two competitions, relative to NCAA D1 = 0. It is added to a line from that competition; negative means an easier competition than D1. Estimator B compares a tournament line with the same player's first D1 season within three years, adjusted for age. Estimator A links competitions through every player seen in more than one, but only 2 of its players link to D1, so its offsets are weakly identified. The Segunda FEB offset compares each bridge player's Segunda FEB line with his D1 season, adjusted for age. None of these offsets enters the walk-forward test.

Competition	Estimator	Offset (BPM)	SE	Players	Status
FIBA U18 AmeriCup 2024	B	−6.26	1.08	19	estimate
Spain's third division (Segunda FEB)	FEB bridge	−1.27	0.61	7	provisional
2026 CAC Games (men)	A	−4.72	3.40	10	provisional
FIBA U17 Centrobasket 2023	A	−9.39	3.86	10	provisional
FIBA U18 AmeriCup 2024	A	−4.14	3.29	10	provisional

Table 3. Standard errors from a player bootstrap. A domestic offset for the Belize league rests on 4 players, below the five-player rule for shared tables, and is not shown; its bridge seasons are all 2026, a test season, so it never entered the walk-forward.

Selection cuts both ways. Players who reach D1 did well enough there to be observed, which makes an offset too generous; players who move after an unusually good line regress afterward, which makes it too harsh. The net direction is untested. An exploratory check on the U18 offset found that players with better tournament lines dropped more by their first D1 season (slope −0.53, 95% interval −0.74 to −0.16), as the second bias predicts for strong lines, but noise in the tournament lines alone produces the same slope. One offset is applied to every position and role, which is also untested (Figure 4).

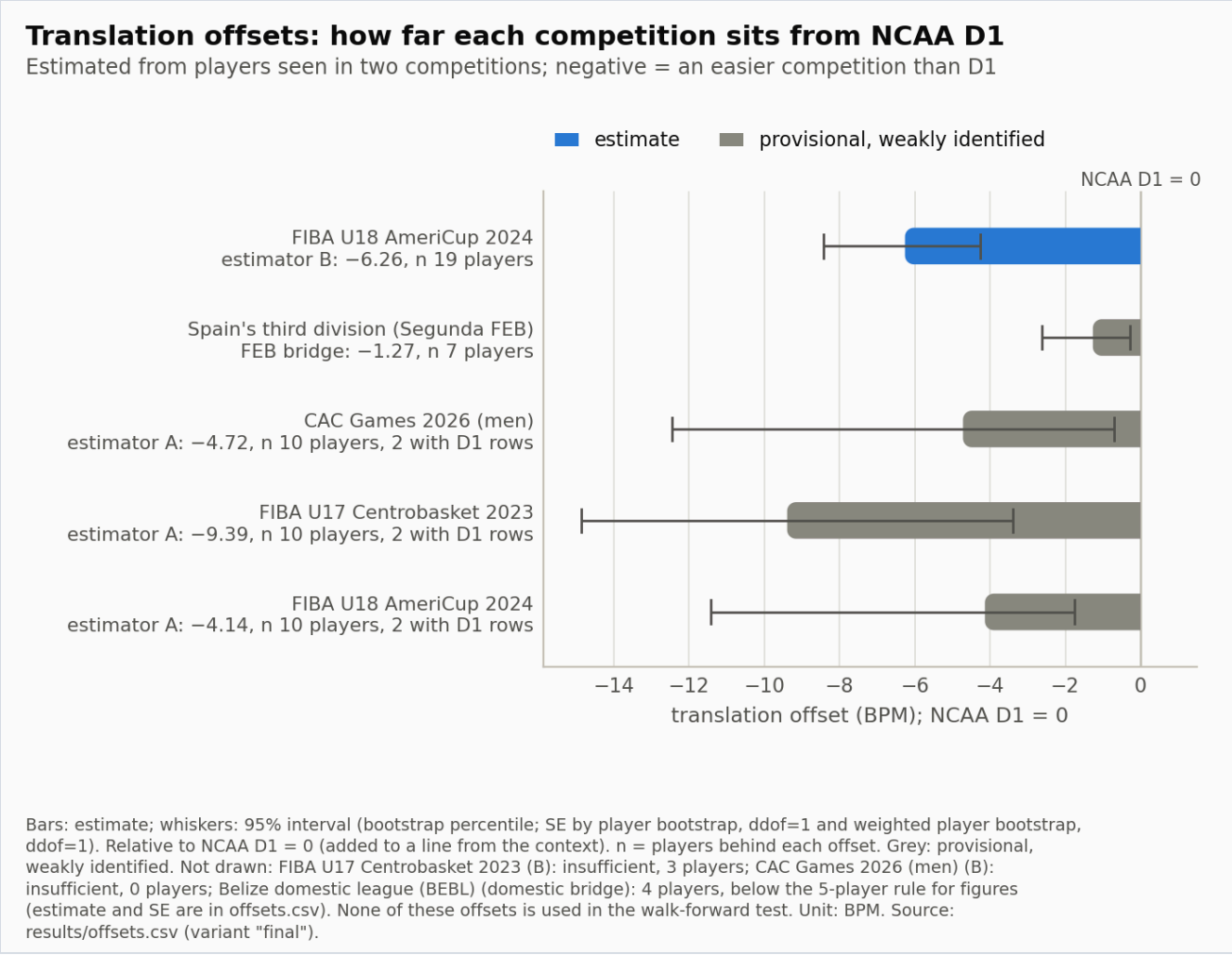


Figure 4. Translation offsets relative to NCAA D1 = 0, with 95% bootstrap intervals and the number of players behind each. Grey: provisional, weakly identified.

5. Checks and diagnostics

- **Leakage test.** A test replaces every later season's rows with random numbers and requires the earlier predictions to stay bit-identical.
- **Second route.** A crosscheck module that imports none of the pipeline's computing code re-derives every walk-forward row, metric, interval, sensitivity subset, residual and offset, and resolves every quoted number to the results cell it repeats.
- **Where the advantage sits.** For the 28 test rows whose training lines all fall inside the D1 input range, the model and the average of covered players tie at 2.68 vs 2.68; for the 74 rows with a line outside it, 2.05 vs 2.57. Of 302 domestic lines, 168 (45% of minutes) have at least one input outside the D1 1st–99th percentile, against 13% of the D1 fitting rows by construction, so for many players the transfer is extrapolating.
- **The transfer inside D1.** Out-of-fold residuals by conference tier run −1.23 for low-major, +0.94 for mid-major and +2.98 for high-major players: the transfer over-predicts low-major players, the group the prior is drawn from.

- **Who is scored.** The 150-minute floor removes 42–46% of each target season's player-seasons, and 17, 18 and 26 players of the previous season do not return. The test speaks for regulars with a track record, not for every player.

6. Data corrections

An independent validation on October 22, 2026 re-derived the inputs on a scratch copy instead of rerunning the shipped code. It found that the walk-forward had scored 2024–26 with shooting-shrinkage constants fit on all four seasons, the test seasons included. The revision refits every constant inside each fold, passes the constants in as arguments, asserts that every training season precedes its target, and adds the leakage test above. The look-ahead was small:

Quantity	As first built (August)	Corrected
Model RMSE	2.213	2.24
Average of covered players, RMSE	2.587	2.60
Raw prior-season average, RMSE	3.331	3.35
50/50 blend, RMSE	2.259	2.28
Gain over the average of covered players	+0.37 (rows resampled)	+0.36 (players resampled)
RMSE with a domestic offset added	2.080	withdrawn

Table 4. The domestic offset behind the withdrawn row was fit on 2026 lines, and 2026 is a test season; no leak-free version exists.

Other corrections: the baseline is now named the average of covered players, because the tables cover only part of each team's minutes; Segunda FEB is Spain's third tier, behind Liga ACB and Primera FEB; the R^2 quoted for domestic lines is the split lane's 0.494, not the full lane's 0.669; four 2026 CAC Games teams whose tables fall short of 200 minutes a game left the pooled tournament shrinkage; and two rows labelled with a season instead of a league were relabelled. Tournament players are now matched to D1 through one registry key per person and the FIBA age-category rule, instead of by name.

7. Limitations

Limit	Mechanism	Likely direction
Scorekeeping	Steals, assists and rebounds are judgment calls, recorded by different scorers in each competition.	Domestic steal rates sit above the D1 range in many lines, pushing composites up; an exploratory scorer-robust lane moves the gain over the covered-player average to −0.34.
Few bridge players	Each offset rests on 4 to 19 players.	Imprecision, not a known direction (standard errors up to 3.86).
Selection of bridge players	Reaching D1 and moving after a good line select in opposite directions.	Net direction untested.
Defense	Box-score BPM under-measures defense, and the outcome shares the blind spot.	Defensive players rated too low; the walk-forward cannot see it.
Pace	Without turnovers or offensive rebounds, possessions cannot be estimated.	If domestic games are faster than D1 games, per-40 rates and composites run high.
Coverage	Players missing from the tables are probably low-minute ones.	The average of covered players overstates the mean of everyone who played.
Transfer fit on D1 only	Untested on players with both domestic and D1 seasons; it over-predicts low-major D1 players.	If Belize players behave like low-major ones, their composites run high.
Choices made while building	Many constants were set while looking at the data.	Intervals look more final than they are; a pre-registered test on 2027 would settle it.
Source rows	The 303 domestic rows pass internal checks but have not been re-checked against their source pages (the original harvest cache is gone).	Unknown until a re-harvest.

8. How a team could use it

The practical lesson is about minutes more than about the scale: a short small-league season should be shrunk hard toward a prior before it is compared with anything, and a plain 50/50 blend gets most of that benefit. The translation offsets are a starting point for placing a FIBA youth or Segunda FEB line beside a D1 line, not a measured exchange rate. One player's next season is uncertain by about ±4.2 BPM, so the rating can help order a group of players but cannot settle a decision about one. The cleanest next test is the 2027 domestic season, with the re-centered prior frozen before it starts.

9. Reproducibility and how this was built

One command, `./run_all.sh`, empties the results and figures folders, rebuilds them from the private, name-free tables in about three minutes, and stops on any failed check: the independent crosscheck, a document check that every quoted number matches the results, a share scan for names, paths, row-level data and small cells, and the test suite (144 tests pass). Two consecutive runs give identical files. The D1 constants are re-derived from the md5-pinned D1 panel by `./run_all.sh --refit-constants` and match the stored values to within 1e-9. The raw pages, the name-to-key registry and every row-level table stay in a private folder outside the repository.

The author chose the question, required a walk-forward test before any rating was trusted, required that everything about players be anonymized, and commissioned the October 22 validation that found the look-ahead. AI coding tools (Claude Code) wrote the code and drafted this paper from the project's results files. The revision applied lessons from Syracuse coursework in statistics, sport analytics, information studies and communication: intervals on differences instead of pass/fail verdicts, a stated decision rule for each test, resampling whole players, sensitivity rows declared before they run, privacy rules for data about minors, and names that say what a number measures. Every change is logged with its evidence in the repository.

Data sources (file dates; access dates not on record)

- Belize domestic league box scores, 2023–2026: basketball.latinbasket.com, August 25, 2026.
- FIBA U17 Centrobasket 2023 and FIBA U18 AmeriCup 2024 box scores: fiba.basketball, August 17, 2026.
- 2026 CAC Games men's basketball box scores: the Games' official results site, August 17, 2026.
- Segunda FEB lines of the bridge players: FEB live stats, August 16, 2026.
- NCAA D1 player panel: barttorvik.com, frozen copy of 90,739 player-seasons.
- FIBA Internal Regulations, Players and Officials, art. 151 (age categories for the estimator-B gate).